

Övningsdugga D02 - Lösningsförslag

1. Realising av konstruktion i VHDL

```
library ieee;
use ieee.std_logic_1164.all;

entity circuit1 is
    port(clock, reset, switch: in std_logic;
         led : out std_logic);
end entity;

architecture behaviour of circuit1 is
    signal switch_prev: std_logic;
    signal switch_edge: std_logic;
    signal led_state : std_logic;

begin
    led      <= led_state;
    switch_edge <= switch and (not switch_prev);

    process(clock, reset) is
    begin
        if (reset = '1') then
            switch_prev <= '0';
        elsif (rising_edge(clock)) then
            switch_prev <= switch;
        end if;
    end process;

    process(clock, reset) is
    begin
        if (reset = '1') then
            led_state <= '0';
        elsif (rising_edge(clock)) then
            if (switch_edge = '1') then
                led_state <= not led_state;
            end if;
        end if;
    end process;
end architecture;
```

